

Name: _____ Partner: _____ Period: _____

Two Robots, One Path

Two robots traverse the same parabolic path in a warehouse.

Robot A: $g_A(t) = (t, t^2)$, $0 \leq t \leq 3$ Robot B: $g_B(t) = (3 - t, (3 - t)^2)$, $0 \leq t \leq 3$

Robot A starts at $(0, 0)$ and Robot B starts at $(3, 9)$. Your goal is to analyze their directions, the path they share, and the slope of the curve between two points.

Part 1 — Tables and Shared Path

1. Complete the table for each robot. Leave no cell blank.

t	Robot A position (x, y) for $g_A(t) = (t, t^2)$	Robot B position (x, y) for $g_B(t) = (3 - t, (3 - t)^2)$
0	_____	_____
1	_____	_____
2	_____	_____
3	_____	_____

2. Do both robots visit the same set of ordered pairs? Explain using your table.

Part 2 — Direction of Motion

3. At $t = 1$, is Robot A moving right or left? Up or down? Show your reasoning by comparing values in the table.

4. At $t = 1$, is Robot B moving right or left? Up or down? Show your reasoning by comparing values in the table.

5. Do the two robots move in the same direction at $t = 1$? Explain what this tells you about the two parametrizations.

Part 3 — Slope of the Curve

6. Using Robot A's data, compute the slope of the curve between $t = 0$ and $t = 2$.

$$x_A(0) = \underline{\hspace{2cm}} \quad x_A(2) = \underline{\hspace{2cm}} \quad \Delta x = \underline{\hspace{2cm}}$$

$$y_A(0) = \underline{\hspace{2cm}} \quad y_A(2) = \underline{\hspace{2cm}} \quad \Delta y = \underline{\hspace{2cm}}$$

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \underline{\hspace{2cm}}$$

7. Using Robot B's data, compute the slope of the curve between the corresponding two points (the same two (x, y) locations). Do you get the same slope? Explain why this makes sense.

Robot B passes through $(0, 0)$ at $t = \underline{\hspace{2cm}}$ and through $(2, 4)$ at $t = \underline{\hspace{2cm}}$.

$$\Delta x = \underline{\hspace{2cm}} \quad \Delta y = \underline{\hspace{2cm}}$$

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \underline{\hspace{2cm}}$$

Part 4 — When Does Each Robot Arrive?

8. At what value of t does Robot A pass through $(1, 1)$? At what value of t does Robot B pass through $(1, 1)$? Show your work (set the component equations equal to 1 and solve).

9. At what value of t does each robot pass through $(0, 0)$? Through $(3, 9)$?
